

CEDLEARN

E-Commerce Purchase Prediction

Capstone Project — Problem Statement

Domain	Level	Type	Dataset
E-Commerce & Retail	Intermediate	Regression	5,055 rows

1. Background

E-commerce platforms generate millions of transactions daily, each influenced by a complex mix of customer demographics, behaviour patterns, and external factors. Understanding what drives a customer’s transaction value is a critical capability for modern retailers — enabling smarter pricing, targeted promotions, and inventory optimisation.

In this project, you will build a machine learning model that predicts the Transaction Amount for an e-commerce order based on features such as customer age, membership history, cart size, discount applied, delivery distance, and contextual factors like season and day of the week.

2. Problem Statement

You are a junior data scientist at an online retail company. The analytics team wants a regression model that can estimate how much a customer is likely to spend on a given transaction. This will support dynamic discount strategies and logistics cost forecasting.

Target Variable: **Transaction_Amount** (continuous, in INR/USD)

3. Dataset Description

File: ecommerce_purchase.csv | Rows: 5,055 | Columns: 12 (11 features + 1 target)

Note: The dataset is intentionally messy. It contains missing values, inconsistent capitalisation in categorical columns (e.g. ‘CREDIT CARD’ vs ‘Credit Card’), implausible numerical values (e.g. negative delivery distances), and outliers in Transaction_Amount. Handling all data quality issues is part of the project.

Feature Name	Type	Description
OrderID	ID	Unique order identifier — drop before modelling

Customer_Age	Numerical	Age of the customer in years
Membership_Years	Numerical	Number of years the customer has been a member
Items_In_Cart	Numerical	Count of items in the order
Discount_Pct	Numerical	Discount applied on the order (%)
Delivery_Distance_KM	Numerical	Distance from warehouse to delivery address (km)
Product_Category	Categorical	Category of product (Electronics, Clothing, Books, etc.)
Payment_Method	Categorical	Mode of payment (Credit Card, UPI, Wallet, Debit Card)
Is_Premium_Member	Binary	Whether the customer holds a premium membership (Yes/No)
Season	Categorical	Season at time of purchase (Summer, Winter, Monsoon, Spring)
Day_Of_Week	Categorical	Day on which the order was placed (Mon–Sun)
Transaction_Amount	TARGET	Total transaction value — the variable to predict

4. Tasks

Complete all tasks below in a Jupyter Notebook. Each task must include code, output, and a brief markdown explanation of your observations.

Type	Task
EDA	Load the dataset. Report shape, data types, and missing values per column.
EDA	Check Payment_Method and Is_Premium_Member for inconsistent capitalisation. List all unique values found before cleaning.
EDA	Plot the distribution of Transaction_Amount. Is it normally distributed or skewed? Discuss implications for regression.
EDA	Plot a boxplot of Transaction_Amount grouped by Product_Category. Which category has the highest median spend?
EDA	Identify physically implausible values in Delivery_Distance_KM (e.g. negative values). How many rows are affected?
EDA	Compute and plot a correlation heatmap for all numerical features. Which feature is most correlated with Transaction_Amount?
Clean	Standardise Payment_Method and Is_Premium_Member to Title Case / consistent format. Verify with unique value counts.

Clean	Replace out-of-range values (e.g. negative Delivery_Distance_KM) with NaN using domain knowledge.
Clean	Impute missing values: median for skewed columns, mean for normal columns, mode for categorical columns.
FE	Create spend_per_item = Transaction_Amount / Items_In_Cart as a derived insight feature.
FE	Create is_weekend = 1 if Day_Of_Week is Sat or Sun, else 0.
FE	Create loyalty_segment: New (<2 years), Regular (2–6 years), Loyal (>6 years). Label encode it.
FE	One-Hot Encode all remaining categorical columns. Explain why OHE is preferred over Label Encoding here.
Model	Split data 80/20 (train/test). Build a Linear Regression and a Random Forest Regressor.
Model	Print MAE, RMSE, and R^2 for each model on the test set. Which model performs better?
Model	Plot Actual vs Predicted values for the best model. How well does it capture high-spend outliers?
Model	Create a comparison table: Model Train R^2 Test R^2 MAE RMSE. Recommend one model for deployment with justification.

5. Deliverables

- Cleaned and feature-engineered dataset saved as a .csv file
- Fully executed Jupyter Notebook (.ipynb) with all code, outputs, and markdown explanations
- At least 6 visualisation plots (EDA + model evaluation)
- Final model comparison table with a written recommendation paragraph